

ShopFlow AWS Project

RDS MySQL Design

Managed Relational Database - MySQL 8.0 - Single-AZ dev

Module 6 of 18

Field	Value
Version	1.0
Date	June 2026
Region	ap-south-1 (Mumbai)
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Application	ShopVerse -- Spring Boot 3.2.5 / Java 21

1. Overview

Amazon RDS provides a managed MySQL 8.0 database for ShopVerse. AWS handles OS patching, engine upgrades, automated daily backups, and hardware failover. ShopVerse connects via JDBC identically to local development -- only the hostname changes. The database resides in private subnets and is inaccessible directly from the internet.

2. Instance Configuration

Setting	Value	Reason
Engine	MySQL 8.0	Matches ShopVerse local development
Instance Class	db.t3.micro	Free-tier eligible (750 hrs/month)
Storage Type	gp3 SSD	Better IOPS than gp2 at same cost
Allocated Storage	20 GB	Minimum for free tier
Max Storage	100 GB	Auto-scaling storage ceiling
Multi-AZ	Disabled (dev)	Enable for production -- 2x cost
Public Access	No	Accessible only from within VPC
DB Name	shopverse	Flyway migrations target this schema
Encryption	AES-256 (at rest)	Always on, no performance cost
Backup Retention	7 days	Point-in-time restore within a week
Performance Insights	Enabled	Identify slow queries -- free 7-day retention

3. MySQL Parameter Group -- shopflow-mysql8-params

Parameter	Value	Purpose
slow_query_log	1	Log queries taking longer than long_query_time
long_query_time	1 (second)	Threshold for slow query
innodb_buffer_pool_size	734 MB	RAM cache for DB pages (~70% of instance RAM)
max_connections	100	Supports multiple app instances
transaction_isolation	READ-COMMITTED	Suitable for CQRS read queries

4. Network Placement

Component	Value
Subnet Group	shopflow-db-subnet-group (prv-1a + prv-1b)
Security Group	shopflow-rds-sg -- allows port 3306 from shopflow-ec2-sg only

Component	Value
VPC	shopflow-vpc (10.0.0.0/16)

5. Credentials

The master username and password are stored in Secrets Manager (/shopflow/db/master). CloudFormation references the secret ARN directly -- the password is never visible in the template. The Spring Boot application connects as a limited-privilege app user (/shopflow/db/app) with SELECT/INSERT/UPDATE/DELETE permissions only, created by Flyway migrations.

6. Spring Boot Connection

Spring Boot Property	Env Variable	Source
spring.datasource.url	DB_HOST, DB_PORT, DB_NAME	SSM Parameter Store
spring.datasource.username	DB_USER	Secrets Manager /shopflow/db/app
spring.datasource.password	DB_PASS	Secrets Manager /shopflow/db/app

7. Cost Estimate

Item	Monthly Cost
db.t3.micro instance (Single-AZ)	\$12.41
20 GB gp3 storage	\$2.30
7-day backup storage	~\$0.00 (free tier)
Total	~\$14.71 / month